# KEFFI AI - CLINICAL DECISION MATRIX & THERAPEUTIC BOUNDARY GUIDELINES

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Project Name: Keffi AI - Clinical Decision Support & AI-Based Therapeutic Support System

Purpose: Comprehensive guide explaining how Keffi AI detects emotional states, selects the correct 10-methodology therapeutic solution, and enforces strict clinical boundaries (Dos & Don'ts).

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## ■ 1. EMOTION DETECTION & INTENT PARSING PIPELINE

Keffi AI uses a 3-tier hybrid detection engine to analyze patient inputs:

1. BERT 96-Emotion Transformer Model:

• Classifies user messages into 96 granular psychiatric states across 11 main categories (Depression, Anxiety, Attrition/Burnout, Trauma, OCD, Bipolar, Psychosis, Substance Use, Eating Disorders, Somatoform, Personality).

2. Cosine Semantic Vector Similarity Router:

• Compares the normalized semantic vector embedding of the user's message against 9 conversational goals (Venting, Advice, Validation, Loneliness, Humor Masking, Distraction, Passive Distress, Factual, Active SOS).

• Normalizes Tanglish slang (e.g. "kaduppa irukku" $\rightarrow$ frustrated, "manda kulambuthu" $\rightarrow$ mentally overwhelmed, "set aagala" $\rightarrow$ disturbed).

3. Psychometric & XAI Scorer (PHQ-9 + SHAP/LIME):

• Maps input tokens to the 9 DSM-5 depression criteria and calculates token-level SHAP risk contribution values.

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## ■■ 2. THE 10 PSYCHOTHERAPEUTIC METHODOLOGIES MATRIX

### WHEN TO APPLY WHAT & WHY

| Method | Clinical Framework | Best Used For (Situations & Emotions) | What Keffi Says / Does |
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| 1 | CBT (Cognitive Behavioral Therapy) | Exam failure, grade panic, catastrophizing, "life is over" thoughts, all-or-nothing thinking. | Identifies specific cognitive distortions. Guides the user to reframe automatic negative thoughts into objective, balanced perspectives. |
| 2 | DBT (Dialectical Behavior Therapy) | Betrayal trauma, explosive rage, intense emotional surges, self-harm impulses. | Applies TIPP crisis skills (Temperature, Paced breathing, Paired muscle relaxation). Focuses on emotional regulation and distress tolerance. |

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| 3 | ACT (Acceptance & Commitment Therapy) | Chronic illness, permanent loss, breakup grief, unwanted intrusive thoughts. | Teaches cognitive defusion—observing painful thoughts as passing clouds without buying into their literal reality. |
| 4 | Somatic Down-Regulation & Grounding | Somatic panic attack, hyperventilation, chest tightness, trembling hands, dizzy panic. | Activates vagus nerve and parasympathetic system via 4-7-8 breathing, 5-4-3-2-1 sensory texture grounding, and body scan. |
| 5 | Rogerian Person-Centered Validation | General sadness, feeling unheard, emotional exhaustion, non-crisis venting. | Provides non-judgmental active listening, warm validation, and emotional mirroring without rushing to fix or advise. |
| 6 | Double-Standard Technique | Self-loathing, perfectionism, harsh inner critic ("I am a failure"). | Highlights self-compassion hypocrisy: asks what they would say to a close friend in the exact same situation. |
| 7 | Micro-Behavioral Activation | Depressive apathy, bed-locking, severe exhaustion, feeling empty. | Offers a 30-second micro-step (taking a sip of cold water, wiggling toes, opening a window) to break behavioral freezing. |
| 8 | Behavioral Experiment & Exposure (NEW) | Social anxiety, fear of judgment, stage fright, dreading upcoming events. | De-catastrophizes worst-case fears by designing a safe micro-experiment testing whether the feared outcome actually happens. |
| 9 | Compassion-Focused Therapy (CFT) (NEW) | Shame, self-blame, feeling unlovable, body image distress. | Activates the parasympathetic soothing-affiliative system through self-soothing touch (hand over heart) and warm self-talk. |
| 10 | Problem-Solving Therapy (PST) (NEW) | Executive function paralysis, multi-task workload overwhelm, project deadlines. | Deconstructs massive workloads and exam schedules into 3 bite-sized, prioritized micro-steps. |

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## ■ 3. WHEN TO OFFER ENTERTAINMENT / DISTRACTION (STORIES, SONGS, PUZZLES)

### RULES FOR ENTERTAINMENT & DISTRACTION

| Solution Tool | When to OFFER IT ■ | When NOT to Offer It ■ (STRICT BANS) |
|---|---|---|
| Calming Music / Ambient Sounds ■ | Mild stress, study anxiety, relaxing before sleep, user requests music/lofi. | STRICT BAN: Active suicidal crisis, acute panic attack (user needs breathing, not music), deep grief. |
| Therapeutic Analogy Stories ■ | Loneliness, searching for meaning, existential confusion, mild depression. | STRICT BAN: Severe agitation, rage, active crisis (user will find long stories frustrating). |

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| Mindful Puzzles & Riddles ■ | Overthinking, mild anxiety distraction, user explicitly asks for distraction/fun. | STRICT BAN: Depressive exhaustion, grief, breakup trauma, exam failure panic (feels insensitive). |

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## ■ 4. THE CLINICAL DOS & DON'TS MATRIX FOR KEFFI AI

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